Ashish Molakalapalli

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EDUCATION

Master of Science in Computer Science *Aug 2025 – May 2027*
Texas A&M University *College Station, TX*

- GPA: 3.833 / 4.0 | Graduating **May 2027** | Coursework: Machine Learning, NLP Foundations & Tech, Artificial Intelligence, Information Storage & Retrieval, Operating Systems, Analysis of Algorithms
- Research focus: LLM Self-Correction and Alignment, Multimodal Learning, Efficient AI Systems, Foundation Models

Bachelor of Technology in Computer Science & Engineering *Dec 2021 – Apr 2025*
IIIT Kottayam *India*

- GPA: 9.34 / 10.0 | Selected Research Scholar, IoT Cloud Research Group – Top 5 of 150+ applicants

INDUSTRY EXPERIENCE

AI for Security Operations Engineer, Intern *May 2025 – Aug 2025*
ServiceNow *San Diego, CA*

- Architected two production LLM agents for automated instance health monitoring and incident investigation across ServiceNow's global security infrastructure, deployed via the Now Assist Skill Kit (JavaScript) and backed by Claude Sonnet on AWS Bedrock.
- Designed a 14-field JSON schema with confidence-based tier escalation (items below 0.85 auto-promote severity) and few-shot prompt strategies ensuring reproducible, structured LLM output at production quality.
- Engineered a 14-tool parallel data-collection layer (GlideRecord, REST callouts, scheduled-job introspection) covering MID server health, certificate expiry, VA scanning, and Tenable hygiene; surfaced real critical findings including a MID server offline since March 2026.
- Delivered automated daily health summaries via Microsoft Teams adaptive-card webhooks, eliminating manual multi-instance review and cutting incident response latency from hours to minutes.

AI Engineer *Sep 2024 – Jul 2025*
Shipsy *Gurugram, India*

- Designed LLM summarization and classification pipelines processing 10K+ support tickets daily for DTDC, integrating Slack, Gmail, and GitHub APIs to automate triage and reduce backlog by 35%.
- Built an enterprise knowledge graph with structured entity linking and RAG over internal documentation, improving query response latency by 40%; won Best Idea Award at Shipsy Hackathon 2025.

Research Intern, Reinforcement Learning *May 2024 – Nov 2024*
CDAC (Centre for Development of Advanced Computing) *Remote*

- Proposed and implemented a Multi-Agent RL scheduler using PPO on simulated HPC workloads, achieving 15% lower energy consumption and 12% faster job makespan versus baseline schedulers.
- Applied centralized training with decentralized execution (CTDE) and cooperative reward shaping; ran ablation studies isolating the contribution of each design choice against independent baselines.

AI/ML Intern *Jun 2024 – Aug 2024*
UST Global *Kochi, India*

- Developed SmartChat, an LLM-powered enterprise chatbot with intent recognition and entity extraction achieving 95% accuracy; fine-tuned GPT-4o on domain-specific data, improving factual correctness by 30% over a zero-shot baseline.
- Authored an internal whitepaper on Enterprise LLM Chatbots establishing prompt engineering and evaluation standards, adopted as team-wide best practices for IT and retail support deployments.

RESEARCH EXPERIENCE

Research Assistant, NLP Lab *Feb 2026 – Present*
Texas A&M University *College Station, TX*

- Constructed a 941-job deep learning runtime benchmark spanning 37 model architectures (CNNs, ViTs, BERT/T5/BART, GPT-2, Llama-2 7B, Qwen2.5, diffusion and video models) across 9 NVIDIA GPU configurations, with a 37-feature schema capturing batch size, sequence length, precision, parallelism, and optimizer configuration.
- Extended the benchmark with a 1,884-run LLM inference dataset capturing prefill/decode split timing, TTFT, TPOT, and KV-cache pressure across 22 models and 4 GPU types.

- Trained GBDT, Random Forest, and neural regressors reaching 23.7% MAPE ($R^2 = 0.881$) on training runtime and 8.8% MAPE ($R^2 = 0.982$) on inference latency; identified log-transform of the target as worth 40–70% relative MAPE improvement.
- Ran out-of-distribution analysis exposing catastrophic cross-architecture failure (CNN to Transformer, 454% MAPE) while per-architecture models held 7–19%, motivating a hybrid analytical-plus-learned correction approach over a purely learned one.
- **Accepted to Findings of EMNLP 2026** – co-authored survey on deep learning job runtime prediction for GPU cluster scheduling, built on the 941-job benchmark above, in collaboration with XPerf and Texas A&M University.

Graduate Researcher, NLP Lab *Sep 2025 – Feb 2026*
Texas A&M University *College Station, TX*

- Built an 8-stage pipeline automating digitization of P&ID engineering schematics into LLM-ready graph structures; fine-tuned YOLOv8 for component detection and applied PaddleOCR with EasyOCR for multi-scale tiled text recognition.
- Applied Hough transforms with morphological operations for line extraction and HDBSCAN with UMAP for unsupervised symbol clustering; emitted structured JSON graphs for downstream LLM reasoning.

PROJECTS

Second Thoughts: When LLM Self-Correction Helps vs Hurts | *Llama-3.1-8B, Mistral-7B, Qwen-2.5-7B, LoRA, GSM8K, HumanEval, MBPP*

- Ran a controlled 3-model $\times$ 5-strategy $\times$ 4-dataset study (96 conditions, 71,595 question-strategy pairs); correction broke 14.7% of correct answers while helping only 4.4% – a 3.4$\times$ damage ratio – with self-consistency dominating correction in 11 of 12 compute-matched comparisons.

Real-Time Job Ingestion and Deduplication Platform | *Python, GitHub Actions, Concurrent I/O, REST APIs, JavaScript*

- Implemented two-pass deduplication (normalized entity matching, then canonical URL matching restricted to job-id-bearing URLs) collapsing 4K raw records into 3.2K unique postings; a self-growing board registry discovers new targets from ingested data with no manual curation.

Financial Document Intelligence: Production RAG System | *RAG, GPT-4o, FAISS, BM25 Hybrid Search, Python, Flask, Docker, AWS EC2*

- Designed a production RAG pipeline over 100+ Schroders financial prospectuses using dense FAISS embeddings with BM25 hybrid re-ranking and GPT-4o generation, improving retrieval precision by 32% over a TF-IDF baseline.

Multimodal Medical AI: Chest X-Ray and Clinical Notes | *ViT-B/16, ClinicalBERT, PyTorch, Cross-Attention, MIMIC-III*

- Ran ablations across early, late, and cross-attention fusion; applied class-weighted loss and threshold calibration for MIMIC-III label imbalance, reducing false negatives on critical findings by 18%.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript (Node.js), R, SQL, CUDA (basic)

AI / ML: PyTorch, TensorFlow, HuggingFace Transformers, JAX (basic), LoRA / PEFT, DeepSpeed ZeRO, Scikit-learn, XGBoost, OpenCV

LLM & Agents: Fine-tuning (SFT, LoRA, QLoRA), Prompt Engineering, Chain-of-Thought, RAG, Tool-use, Agentic Pipelines, AWS Bedrock, OpenAI API, Anthropic API

Evaluation & Alignment: Self-Correction Analysis, Per-instance Correction Matrix, Ablation Design, Reward Modeling, ROUGE, BERTScore, Hallucination Detection

MLOps: AWS (EC2, S3, Bedrock, SageMaker), Docker, Git, FAISS, SLURM, Weights & Biases, Experiment Tracking, REST APIs

Research Areas: LLM Alignment, Foundation Models, Multimodal Learning, Efficient Inference (Quantization, LoRA), Agentic AI, NLP

ACHIEVEMENTS & PUBLICATIONS

- **Findings of EMNLP 2026 (Accepted)** – *Deep Learning Job Runtime Prediction for GPU Cluster Scheduling*; co-authored survey on deep learning job runtime prediction for GPU cluster scheduling, in collaboration with XPerf and Texas A&M University, grounded in a 941-job, 37-feature benchmark across 9 GPU configurations plus a 1,884-run inference dataset.
- **Second Thoughts: When LLM Self-Correction Helps vs Hurts** – Course paper (CSCE 638, NLP), Texas A&M; 96-condition study across 71,595 pairs, novel D1/D2 SFT datasets with DeepSeek-R1 attribution-conditioned LoRA fine-tuning, +17.82 GSM8K gain on Mistral-7B.
- **Hackathon Winner** – Best Idea Award, Shipsy Hackathon 2025 (40+ teams) for an autonomous standup-to-Jira pipeline; National Finalist, Smart India Hackathon competing against 500K+ participants across institute, regional, and national rounds.
- **AI/ML Club Sub-Lead**, IIIT Kottayam (2024) – Directed 8 ML workshops and 3 hackathons; grew active membership by 40% and mentored 30+ students. **IEEE Event Organizer** (2023) – Hosted the IoT Cloud Research Symposium with 200+ attendees across 6 research sessions.